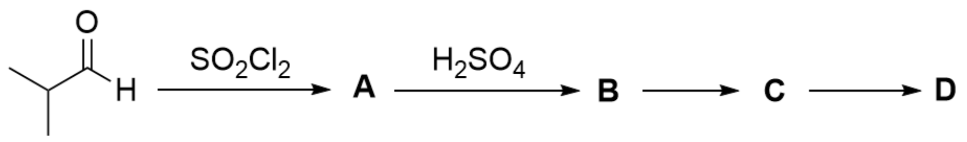
Question



The image depicts an organic tandem reaction: CC(C)C([H])=O>O=S(Cl)(Cl)=O>[*],[*]>O=S(O)(O)=O>[B],[B}>[C],[C]>[D].

Known information about the reaction in the image:

1. **A** contains a carbonyl group.

2. Reaction conditions for **B** to **C**: (a) $\mathrm{NaOH, 185^{\circ}C}$; (b) $\mathrm{t-BuOK, 75^{\circ}C}$.

3. Reaction conditions for **C** to **D**: (c) $\mathrm{O_3, DCM, MeOH, -78^{\circ}C}$; (d) $\mathrm{O_3, DCM, -78^{\circ}C; then\ Me_2S}$.

4. $1\ \mathrm{mol}$ of **D** decomposes above $-40^{\circ}\mathrm{C}$ to yield a single gaseous product, with a volume of approximately $74\ \mathrm{L}$ at room temperature and atmospheric pressure, and a density of about $1.8\ \mathrm{kg/m^3}$.

Which of the following statements about the unknowns **A-D** is correct?

A. **A** is acidic

B. **C** The average molecular weight of the completely hydrolyzed product does not exceed 85.

C. There are two chemical environments for hydrogen atoms in **D**.

D. To achieve a higher reaction yield for the conversion of **B** to **C**, reaction condition (a) should be selected.

E. In order to achieve a higher reaction yield for the generation of **D** from **C**, reaction condition (d) should be selected.

F. None of the above statements are correct

Answer

Correct Answer: F

Detailed Explanation

The substrate is an aldehyde. Since **A** contains a carbonyl group, the aldehyde functional group should be retained. Thionyl chloride serves as the chlorinating reagent, leading to α-halogenation of the aldehyde, and the product is CC(C)(Cl)C([H])=O. This compound lacks α-hydrogens, and the aldehyde hydrogen is difficult to dissociate, rendering it essentially non-acidic. Option A is incorrect.

CHECKPOINT 1 PTS

α-Halogenation of the aldehyde occurs

CHECKPOINT 1 PTS

A is CC(C)(Cl)C([H])=O

The reaction from **A** to **B** is relatively difficult to deduce, so we start with the decomposition of **D**. Based on the fact that **D** decomposes to produce only one gas, and using the ideal gas law, the properties of the gas can be determined:

$$\mathrm{n = pV/RT = 3.03\ mol}$$

$$\mathrm{M} = \rho \mathrm{V_m} = 44.01$$

This indicates that the gas is carbon dioxide, and $3\ \mathrm{mol}$ is produced. Therefore, the chemical formula of **D** is inferred to be C_3O_6, and its structure can only be O=C(OC(O1)=O)OC1=O. This structure contains no hydrogen atoms, so option C is incorrect.

CHECKPOINT 1 PTS

D decomposes to produce only carbon dioxide, generating $3\ \mathrm{mol}$

CHECKPOINT 1 PTS

The chemical formula of **D** is C_3O_6, and its structure can only be O=C(OC(O1)=O)OC1=O

The conversion of **C** to **D** is clearly an ozonolysis reaction of a double bond, where the double bond cleaves to form carbonyl groups. Thus, the structure of **C** can be deduced as C/C(C)=C(O/C(O/1)=C(C)\C)/OC1=C(C)/C. Complete hydrolysis of this structure yields only CC(C(O)=O)C, with a molecular weight of 88, so option B is incorrect.

CHECKPOINT 1 PTS

The structure of **C** is C/C(C)=C(O/C(O/1)=C(C)\C)/OC1=C(C)/C

CHECKPOINT 1 PTS

Hydrolysis of **C** yields only CC(C(O)=O)C

The conversion of **B** to **C** occurs under strong base conditions. Based on the double-bond structure of **C**, this is likely an elimination reaction. Combining this with the structure of **A**, it can be inferred that the conversion of **A** to **B** is a trimerization reaction. Under acidic conditions, the carbonyl group is protonated and then attacked by another carbonyl molecule, leading to trimerization. Thus, **B** is CC(C)(Cl)C1OC(C(C)(Cl)C)OC(C(C)(Cl)C)O1.

CHECKPOINT 1 PTS

B is CC(C)(Cl)C1OC(C(C)(Cl)C)OC(C(C)(Cl)C)O1

The conversion of **B** to **C** is an elimination reaction. However, under high-temperature, strongly basic conditions, hydroxide ions act as strong nucleophiles, making the six-membered ring prone to hydrolysis. Therefore, using a less nucleophilic base like potassium tert-butoxide results in higher yields, so option D is incorrect.

CHECKPOINT 1 PTS

The conversion of **B** to **C** is an elimination reaction; using a less nucleophilic base like potassium tert-butoxide yields higher efficiency

The conversion of **C** to **D** involves ozonolysis of the double bond. If dimethyl sulfide is used to treat the resulting ozonide intermediate, the strong nucleophilicity of the sulfide may open the six-membered ring. Hence, reaction condition (c) is preferable, so option E is incorrect.

CHECKPOINT 1 PTS

The conversion of **C** to **D** is an ozonolysis reaction; the strong nucleophilicity of the sulfide may open the six-membered ring

In conclusion, options A-E are all incorrect, and option F is correct.